Secure Control for T–S Fuzzy Wind Turbine Systems Under Hybrid Cyberattacks via an Adaptive Memory Event-Triggered Mechanism

Dong Xu, Yajuan Liu, and Sangmoon Lee, *Member, IEEE*

***Abstract*—This article addresses the secure control problem for the Takagi–Sugeno (T–S) fuzzy wind turbine system (WTS) subject to hybrid cyberattacks. To reduce system performance loss and redundant data transmission under such attacks, a novel adaptive memory event-triggered mechanism (ETM) is proposed, offering two key advantages. First, unlike traditional ETMs that rely solely on current system information, the adaptive memory ETM utilizes historical release data to adjust communication frequency and enhance control effectiveness. Second, an attack-related triggering condition and adaptive law are introduced to reduce the abnormal-data transmission induced by denial of service (DoS) attacks and extend the lifespan of the sensor node. Sufficient conditions are derived to ensure the mean-square $\mathcal{H}_\infty$ asymptotic stability of the resulting T–S fuzzy WTS, while also guaranteeing uniformly ultimate boundedness in the presence of DoS attacks. Finally, an illustrative example is used to show the effectiveness of the proposed method.**

***Index Terms*—Adaptive memory event-triggered mechanism (AMETM), hybrid cyberattacks, secure control, wind turbine system (WTS).**

I. Introduction

THE global population growth and increasing manufacturing demands lead to a continuous rise in energy demand, and conventional, nonrenewable fossil fuels will be inadequate to meet future energy needs. The development and application of alternative energy sources are crucial for the sustainable progression of contemporary society. As a significant alternative to fossil fuels, wind energy has gained substantial attention [1], [2], [3]. Recently, several nations have fostered actively the development of wind energy-dependent industries. The research on the stability of wind turbine systems (WTSs) has, therefore, received wide attention [4], [5], [6]. Among various WTS generators, the permanent magnet synchronous generator (PMSG) is favored for its direct drive capability, low rotational speed, and reduced maintenance costs. However, WTS models employing PMSG encounter inevitably difficulties caused by nonlinearity, which brings challenges in stability analysis and control synthesis. The Takagi–Sugeno (T–S) fuzzy method shows its merit in dealing with these nonlinear complexities. By employing suitable membership functions, this approach can transform nonlinear systems into a sequence of linear subsystems, which has attracted many researchers on T–S fuzzy PMSG-based WTSs, see, for instance, [7], [8], [9] and the references therein.

In addition, with the fast development of wireless communication technology, networked control has been proposed in the field of WTS control [10], [11], [12], [13]. In comparison with conventional control methods, the networked control approach provides the benefits of remote operation and easy scalability [14]. Nonetheless, open-communication networks may render systems vulnerable to cyberattacks. Broadly speaking, cyberattacks can be categorized into two primary types: denial-of-service (DoS) and deception attacks. DoS attacks occupy the limited communication bandwidth with abnormal requests, disrupting the transmission of data packets. Deception attacks compromise data integrity by hijacking sensors and altering control signals. In scenarios of both attacks occurring simultaneously, the system dynamics may experience significant damage, resulting in malfunction. Recent research has increasingly focused on addressing the security challenges of networked control in the presence of hybrid cyberattacks [15], [16], [17], [18]. However, to the best of authors' knowledge, no research has been conducted so far to specifically address the security control issue in PMSG-based WTSs when hybrid cyberattacks occur.

Another worth noting point is that the communication resources of networked WTSs are inherently limited. How to enhance communication efficiency without compromising the system performance remains a critical challenge. Traditional time-triggered control strategies, which need the periodic measurement and transmission of data packets, lead to unnecessary consumption of communication resources. In response to this inefficiency, the introduction of the event-triggered mechanism (ETM) offers a feasible solution [19]. ETMs allocate communication resources according to predefined, event-triggered criteria, transmitting only signals that meet specific triggering conditions while omitting those that do not. Over the past two decades, various types of ETMs have been investigated. This evolution began with the early continuous ETM [19] and has expanded to different ETMs like discrete ETMs [20], distributed

Received 13 July 2024; revised 30 September 2024; accepted 30 December 2024. Date of publication 3 January 2025; date of current version 5 May 2025. This work was supported in part by the National Natural Science Fund of China under Grant 62273144 and in part by the National Research Foundation of Korea (NRF) through the Korea government, and in part by the Ministry of Science and ICT under Grant RS-2024-00350118. Recommended by Associate Editor H. Schulte. *(Corresponding authors: Yajuan Liu; Sangmoon Lee.)*

Dong Xu and Yajuan Liu are with the School of Control and Computer Engineering, North China Electric Power University, Beijing 102206, China (e-mail: dxu1997@126.com; yajuan.liu.12@gmail.com).

Sangmoon Lee is with the School of Electronic and Electrical Engineering, Kyungpook National University, Daegu 41566, South Korea (e-mail: moony@knu.ac.kr).

Digital Object Identifier 10.1109/TFUZZ.2025.3525778

1941-0034 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Institut Teknologi Sepuluh Nopember. Downloaded on March 29,2026 at 17:29:57 UTC from IEEE Xplore. Restrictions apply.

ETMs [21], dynamic ETMs [22], adaptive ETMs [23], and memory ETMs [24], among others. To present, the application of ETMs in WTSs has been discussed in papers [25] and [26]. By utilizing the classic discrete ETM and the T–S fuzzy method, the paper [25] discussed the stability problem of PMSG-based WTSs with back-to-back converters. In [26], an adaptive memory ETM was designed to investigate the output feedback stabilization problem for T–S fuzzy WTSs with uncertain parameters. However, the aforementioned ETMs possess their own limitations. The discrete ETM in [25] uses a static triggering threshold, failing to reflect the system's dynamic nature, while the adaptive memory ETM in [26] requires additional hardware for continuous assessment and computation of triggering conditions. Furthermore, many ETMs neglect the impact of cyberattacks on WTSs. For instance, deception attacks that tamper with control signals may lead to significant fluctuations in system states. In such circumstances, it is required to reduce the difficulty of the triggering condition and release more data packets to suppress these fluctuations. Conversely, when DoS attacks disrupt data transmission, no matter how many data packets are released, control will lose its effect. Therefore, it is more sensible to diminish the probability of releasing packets to prolong the battery life of the nodes.

Motivated by the aforementioned problems, in this article, we investigate the event-triggered security control for WTSs under hybrid cyberattacks and propose an attack-related adaptive memory event-triggered mechanism (AMETM). The term "memory" in this mechanism has a different implication from its usage in [26], where it denotes the storage of continuously weighted average output signals to reduce false triggers caused by external disturbances. The "memory" in this study refers to the storage of a set number of recently released signals. By employing a weighted error between these signals and the current sampling signal, one can increase the probability of releasing packets at the peaks (troughs) of the state curve, which speeds up system convergence during significant fluctuations. Moreover, the attack-related event-triggered condition and adaptive law can adjust dynamically the communication frequency according to the presence or absence of DoS attacks. This mitigates the impact of such attacks and enhances communication efficiency. The main contributions of this article are as follows.

a) An attack model of the T–S fuzzy PMSG-based WTS is established for the first time, which includes hybrid cyberattacks, network-induced delays, asynchronous premise variables, and event-triggered communication mechanisms within a unified framework.
b) A novel attack-related AMETM is designed to improve communication efficiency and alleviate the detrimental impacts of hybrid cyberattacks. Unlike traditional memory ETMs, this mechanism allows for dynamic adjustment of communication frequency based on the occurrence of DoS attacks. Moreover, a novel adaptive law, which is based on historical release signals and cyberattacks, is formulated to adjust flexibly the triggering threshold.
c) A memory control design method is proposed to derive simultaneously the memory-based output feedback control gains and triggering parameters.

Sufficient conditions are presented to guarantee that the PMSG-based WTS achieves mean-square $\mathcal{H}_\infty$ asymptotic stability in the absence of DoS attacks, and uniform ultimate boundedness in the presence of DoS attacks.

The rest of this article is organized as follows. Section II formulates the problem. The main results are stated in Section III. A numerical example is presented in Section IV. Finally, Section V concludes this article.

Notation: $\mathbb{N}$ symbolizes the set of natural numbers. M^T represents the transpose of matrix M. I and 0 correspond to the identity matrix and the zero matrix with appropriate dimensions, respectively. diag$\{\cdot\}$ stands for a diagonal matrix. $\otimes$ denotes the Kronecker product. $||\cdot||$ and $||\cdot||_2$ denote the Euclidean norm for vectors and the 2-norm for matrices, respectively. $\mathbb{E}\{\cdot\}$ indicates the expectation of a stochastic variable.

II. System Formulation

In this section, a T–S fuzzy system model of a WTS with a nonlinear PMSG is developed by taking communication efficiency and security into account.

A. T–S Fuzzy Model of PMSG-Based WTS

Referring to [27], we establish the mathematical model for the surface-mounted PMSG using the dq reference frame voltage equations and the generator torque $\mathbb{T}_e(\cdot)$, described by

$$\begin{cases} \mathbb{V}_{sd}(t) = \mathbb{r}_s \mathbb{i}_{sd}(t) + \mathbb{L}_d \frac{d\mathbb{i}_{sd}(t)}{dt} - \omega_e(t)\mathbb{L}_q \mathbb{i}_{sq}(t) \\ \mathbb{V}_{sq}(t) = \mathbb{r}_s \mathbb{i}_{sq}(t) + \mathbb{L}_q \frac{d\mathbb{i}_{sq}(t)}{dt} + \omega_e(t)\mathbb{L}_d \mathbb{i}_{sd}(t) \\ \qquad\qquad + \omega_e(t)\vartheta_f \\ \mathbb{T}_e(t) = \frac{3}{2}\mathbb{m}_p \mathbb{i}_{sq}(t)\vartheta_f \end{cases} \tag{1}$$

where $\mathbb{V}_{sd}(t)$, $\mathbb{V}_{sq}(t)$, $\mathbb{i}_{sd}(t)$, and $\mathbb{i}_{sq}(t)$ represent the stator voltages and currents in the dq reference frame at time t, respectively. $\omega_e(t)$ represents the rotor electrical angular speed with $\omega_e(t) = \mathbb{m}_p \omega_g(t)$, where $\omega_g(t)$ is the rotor angular speed.

Moreover, the mechanical behavior of the wind turbine equipped with a PMSG is governed by the following equation:

$$\mathbb{I}\frac{d\omega_g(t)}{dt} = \mathbb{T}_e(t) - \mathbb{T}_m(t) - \mathbb{F}_c \omega_g(t). \tag{2}$$

The aerodynamic torque, $\mathbb{T}_m(t)$, and the tip speed ratio, $\mu(t)$, are specified as follows:

$$\mathbb{T}_m(t) = \frac{\alpha\pi\mathbb{R}^3 v^2(t)\mathcal{C}_P(\mu(t), \tilde{\beta})}{2\mu(t)} \tag{3}$$

where

$$\mathcal{C}_P(\mu(t), \tilde{\beta}) = 0.5109\left(\frac{116}{\tilde{\mu}(t)} - 0.4\tilde{\beta} - 5\right)e^{\frac{-21}{\tilde{\mu}(t)}}$$

$$\tilde{\mu}(t) = \frac{1}{\mu(t) + 0.08\tilde{\beta}} - \frac{0.035}{\tilde{\beta}^3 + 1}, \quad \mu(t) = \frac{\mathbb{R}\omega_g(t)}{v(t)}.$$

Herein, $\mathcal{C}_P(\mu(t), \tilde{\beta})$ represents the wind turbine's power coefficient with $\tilde{\beta}$ and $\mu(t)$ denoting the blade pitch angle and tip speed ratio, respectively. The physical meanings of other parameters are given in Table I, and the detailed construction of

Authorized licensed use limited to: Institut Teknologi Sepuluh Nopember. Downloaded on March 29,2026 at 17:29:57 UTC from IEEE Xplore. Restrictions apply.

TABLE I
PARAMETERS IN THE PMSG-BASED WTS MODEL

Parameters	Physical meanings	Values
$\mathbb{m}_p$	Number of poles	10
$\mathbb{r}_s$	Stator resistance	$0.4\,\Omega$
$\mathbb{L}_d$, $\mathbb{L}_q$	Stator inductance	17.5 mH
$\mathbb{F}_c$	Friction coefficient	0.009 N.m.s/rad
$\mathbb{I}$	Moment of inertia	0.188 kg/m^2
$\mathbb{R}$	Blade Length	1.94 m
α	Air density	1.25 kg/m^3
ϑ_f	Magnetic flux linkage	0.445 wb
μ_{opt}	Tip speed ratio	8.1
$\mathcal{C}_{p\max}$	Maximum power coefficient	0.42
$v(t)$	Wind velocity	12 m/s

the referenced PMSG-based WTS model (1)–(3) can be found in [8] and [26].

Set $x(t) = [\omega_g(t)\ \mathbb{i}_{sq}(t)\ \mathbb{i}_{sd}(t)]^T$, $u(t) = [\mathbb{V}_{sq}(t)\ \mathbb{V}_{sd}(t)]^T$ as the state and control input, respectively, We formulate a state-space of WTS as

$$\begin{cases} \dot{x}(t) = A(t)x(t) + Bu(t) + F(t)\varpi(t) \\ y(t) = Cx(t) \end{cases} \tag{4}$$

where $y(\cdot)$ represents the measured output and $\varpi(t)$ represents the external disturbance belonging to $\mathcal{L}_2[0,\infty)$. Subsequently, the system matrices can be expressed as

$$A(t) = \begin{bmatrix} \frac{-\mathbb{F}_c}{\mathbb{I}} & \frac{3\mathbb{m}_p\vartheta_f}{2J} & 0 \\ -\frac{\mathbb{m}_p\vartheta_f}{\mathbb{L}_q} & -\frac{\mathbb{r}_s}{\mathbb{L}_q} & -\frac{\mathbb{m}_p\omega_g(t)\mathbb{L}_d}{\mathbb{L}_q} \\ 0 & \frac{\mathbb{m}_p\omega_g(t)\mathbb{L}_q}{\mathbb{L}_d} & -\frac{\mathbb{r}_s}{\mathbb{L}_d} \end{bmatrix}$$

$$B = \begin{bmatrix} 0 & 0 \\ \frac{1}{\mathbb{L}_q} & 0 \\ 0 & \frac{1}{\mathbb{L}_d} \end{bmatrix} \quad F(t) = \begin{bmatrix} \frac{-\mathbb{T}_0\omega_g(t)}{\mathbb{I}} \\ 0 \\ 0 \end{bmatrix}$$

$$C = \begin{bmatrix} 0 & 1 & 1 \end{bmatrix} \quad \mathbb{T}_0 = \frac{\alpha\pi\mathbb{C}_{p\max}\mathbb{R}^5}{2\mu_{\text{opt}}^3}.$$

It is noted that $A(t)$ and $F(t)$ are state-dependent. In order to address the nonlinear complexity in system (4), the T–S fuzzy method can be utilized as follows.

Plant Rule i: IF $w_g(t)$ is ψ_i, then

$$\begin{cases} \dot{x}(t) = A_ix(t) + Bu(t) + F_i\varpi(t) \\ y(t) = Cx(t) \end{cases} \tag{5}$$

where $\omega_g(t) = x_1(t) \in [x_{1\min}\ x_{1\max}] = [\mathbb{D}_1\ \mathbb{D}_2]$ is set as the premise variable, ψ_i^s, $s \in \{1,\dots,p\}$, which represents the fuzzy set for ith rule, $i \in \{1,\dots,q\}$, $p = 1$ and $q = 2$ corresponding to the number of premise variables and fuzzy rules, respectively, and

$$A_1 = \begin{bmatrix} \frac{-\mathbb{F}_c}{\mathbb{I}} & \frac{3\mathbb{m}_p\vartheta_f}{2J} & 0 \\ -\frac{\mathbb{m}_p\vartheta_f}{\mathbb{L}_q} & -\frac{\mathbb{r}_s}{\mathbb{L}_q} & -\frac{\mathbb{m}_p\mathbb{D}_1\mathbb{L}_d}{\mathbb{L}_q} \\ 0 & \frac{\mathbb{m}_p\mathbb{D}_1\mathbb{L}_q}{\mathbb{L}_d} & -\frac{\mathbb{r}_s}{\mathbb{L}_d} \end{bmatrix} F_1 = \begin{bmatrix} \frac{-\mathbb{T}_0\mathbb{D}_1}{\mathbb{I}} \\ 0 \\ 0 \end{bmatrix}$$

$$A_2 = \begin{bmatrix} \frac{-\mathbb{F}_c}{\mathbb{I}} & \frac{3\mathbb{m}_p\vartheta_f}{2J} & 0 \\ -\frac{\mathbb{m}_p\vartheta_f}{\mathbb{L}_q} & -\frac{\mathbb{r}_s}{\mathbb{L}_q} & -\frac{\mathbb{m}_p\mathbb{D}_2\mathbb{L}_d}{\mathbb{L}_q} \\ 0 & \frac{\mathbb{m}_p\mathbb{D}_2\mathbb{L}_q}{\mathbb{L}_d} & -\frac{\mathbb{r}_s}{\mathbb{L}_d} \end{bmatrix} F_2 = \begin{bmatrix} \frac{-\mathbb{T}_0\mathbb{D}_2}{\mathbb{I}} \\ 0 \\ 0 \end{bmatrix}.$$

In this way, the T–S fuzzy WTS with surface-mounted PMSG can be described as follows:

$$\begin{cases} \dot{x}(t) = \sum_{i=1}^{2}\theta_i(t)(A_ix(t) + Bu(t) + F_i\varpi(t)) \\ y(t) = Cx(t) \end{cases} \tag{6}$$

where $\theta_i(t)$ represents the normalized membership function and satisfies to the following condition:

$$\theta_1(t) = \frac{\mathbb{D}_2 - x_1(t)}{\mathbb{D}_2 - \mathbb{D}_1},\ \theta_2(t) = 1 - \theta_1(t)$$

$$\sum_{i=1}^{2}\theta_i(t) = 1,\ x_1(t) = \theta_1(t)\mathbb{D}_1 + \theta_2(t)\mathbb{D}_2.$$

B. AMETM Without DoS Attacks

Assumption 1: The set sampling period is denoted as h. The sensor operates in a time-triggered manner while the controller functions in an event-triggered manner. $\Phi_1 = \{0, h, \dots, kh, \dots\}, k \in \mathbb{N}$, represents the sampling instant sequence. $\Phi_2 = \{s_0h, s_1h, \dots, s_kh, \dots\}, s_k \in \mathbb{N}$, represents the release instant sequence, and $\Phi_2 \subset \Phi_1$.

Remark 1: As noted in [20], the ETM that utilizes time-triggered sensors assesses the triggering condition at fixed sampling instants. This mechanism not only prevents the Zeno phenomenon but also eliminates the need for additional equipment to continuously monitor the measurement signal. Consequently, it significantly reduces the complexity and cost of implementing an event-triggered controller.

By utilizing a series of sampling signals at historical release instants in Φ_2, we introduce a novel AMETM. Define $\bar{x}(t) = (1/m)\sum_{r=1}^{m}x(s_{k-r+1}h)$, $e_r(t) = x(s_kh + lh) - x(s_{k-r+1}h), r, l \in \mathbb{N}$, the event-triggered condition is expressed as

$$s_{k+1}h = s_kh + \inf\left\{lh \middle| \sum_{r=1}^{m}\lambda_re_r^T(t)\Theta e_r(t) > \varsigma(s_kh)\bar{x}^T(t)\Theta\bar{x}(t)\right\} \tag{7}$$

where $\Theta = C^TWC$ and $W > 0$ denotes the triggering matrix to be designed, m represents the number of recently released signals stored by the buffer, $\lambda_r \in [0,1]$ denote the weighted parameters, and $\sum_{r=1}^{m}\lambda_r = 1$. $\varsigma(s_kh)$ represents the measurement of the dynamic threshold $\varsigma(t)$ at $t = s_kh$. This value can be

Authorized licensed use limited to: Institut Teknologi Sepuluh Nopember. Downloaded on March 29,2026 at 17:29:57 UTC from IEEE Xplore. Restrictions apply.

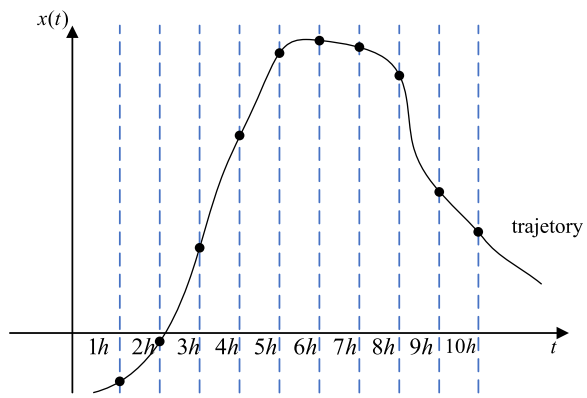

Fig. 1. Schematic for the sampling signals of the controlled system.

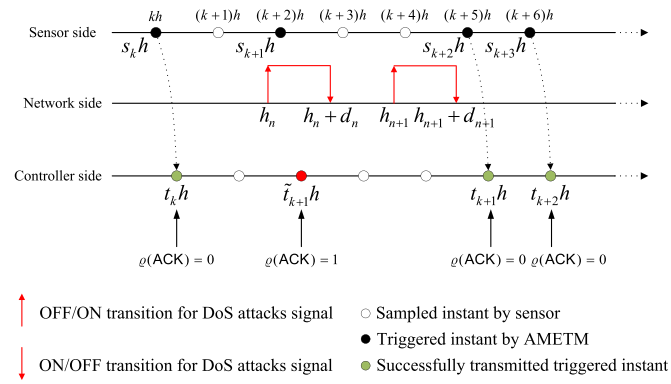

Fig. 2. Timing diagram for AMETM under DoS attacks.

regulated through the following memory-based adaptive law:

$$\dot{\varsigma}(t) = \left(\frac{1}{\varsigma^2(t)} - \frac{\kappa}{\varsigma(t)}\right) \sum_{r=1}^{m} \lambda_r e_r^T(t) \Theta e_r(t) \tag{8}$$

where $\varsigma(t) > 0$ and the constant $\kappa > 0$ are employed to regulate the convergence speed of the threshold function.

Remark 2: In memoryless ETMs [28], [29], [30], the triggering conditions rely solely on the error between the current and most recently released signals. While these mechanisms appear reasonable—since a smaller error suggests the newly sampling signal is less likely to be released—it does not always hold. For instance, as illustrated in Fig. 1 , under the triggering conditions of memoryless ETMs, a small error between $x(6h) - x(5h)$ and $x(7h) - x(6h)$ can prevent the release of these sampling signals, despite their values being significantly larger than others. Releasing these signals would effectively reduce the system's transient response time. Therefore, this study employs a set of historical release data to adjust the release probability between adjacent sampling moments.

Remark 3: Different from the memory ETM in [24] that utilized a fixed threshold, this research proposed an AMETM with a memory-based adaptive law. This adaptive law can regulate dynamically the triggering threshold according to the weighted error, which therefore allows a more rational adjustment of the communication frequency.

C. Energy-Constrained DoS Attacks

Assumption 2: The DoS attacks concerned in this article are energy-constrained [31] rather than stochastic distribution [32] or periodic [33]. Moreover, these DoS attacks are specifically targeted at the communication between the sensor and controller.

Under DoS attacks, data packets are unable to transmit as expected, preventing the controller from acquiring the necessary released signals, timely updates to the controller and potentially resulting in degraded system performance. The sequence of DoS attacks, from activation to termination, can be represented as $\mathbb{H}_n = \{\mathbb{h}_n\} \cup (\mathbb{h}_n, \mathbb{h}_n + \mathbb{d}_n]$, where $\mathbb{h}_n > 0$ and $\mathbb{d}_n \geq 0$ represents the duration of these attacks. Then, the DoS attacks interval in $[0, t)$ can be uniformly expressed as $\mathbb{H}(0, t) = \bigcup_{n \in \mathbb{N}} \mathbb{H}_n$.

Defining $\Phi_3 = \{t_0 h, t_1 h, \ldots, t_k h, \ldots\}$ as the set of successfully releasing instants, it is evident that $\Phi_3 \subseteq \Phi_2 \subseteq \Phi_1$. Notably, in the absence of DoS attacks, each released signal can be transmitted successfully, which implies that $\Phi_2 = \Phi_3$.

In order to characterize the negative effects of energy-constrained DoS attacks, for every successfully releasing time interval $[t_k h, t_{k+1} h)$, $k \in \mathbb{N}$, there are two cases.

Cases 1: if $[t_k h, t_{k+1} h) \cap \mathbb{H}(0, t) = \varnothing$, then no DoS attacks happen within the releasing time interval $[t_k h, t_{k+1} h)$, such as the interval $[t_{k+1} h, t_{k+2} h)$ shown in Fig. 2.

Cases 2: if $[t_k h, t_{k+1} h) \cap \mathbb{H}(0, t) \neq \varnothing$, then DoS attacks are present during the releasing time interval $[t_k h, t_{k+1} h)$, such as the interval $[t_k h, t_{k+1} h)$ shown in Fig. 2.

The acknowledgment (ACK) character technique was utilized to determine the success of data transmission in [31], where the ACK signal is prioritized highest and experiences minimal transmission delay. Upon receiving error-free data, the controller provides a correct ACK signal (i.e., $\varrho(\mathsf{ACK}) = 0$), which indicates successful transmission and readiness for the next. Conversely, if a character error is detected, the ETM will not receive a correct ACK signal and will reinitiate data transmission until a correct ACK is received.

Specifically, an ACK signal of $\varrho(\mathsf{ACK}) = 0$, corresponding to case 1, denotes the absence of DoS attacks at the triggering instant, permitting communication with the controller. In contrast, $\varrho(\mathsf{ACK}) = 1$ indicates communication obstruction by DoS attacks, which is in line with case 2. Consequently, the ACK signal $\varrho(\mathsf{ACK})$ can be characterized as DoS attack behavior, that is

$$\varrho(\mathsf{ACK}) = \begin{cases} 0, & \text{no DoS attacks} \Leftrightarrow \text{Case 1} \\ 1, & \text{under DoS attacks} \Leftrightarrow \text{Case 2.} \end{cases} \tag{9}$$

Remark 4: It is noteworthy that the wireless communication module of the sensor node consumes the most energy during transmission [34]. However, in the event of DoS attacks, none of the transmitted packets reach their destination, resulting in unnecessary energy consumption. In such cases, deferring the release of sampling packets until the DoS attack ceases is a pragmatic strategy. Nevertheless, traditional ETMs are unable to detect DoS attacks, leading to ongoing, futile attempts at data transmission. The introduction of the ACK technology facilitates

Authorized licensed use limited to: Institut Teknologi Sepuluh Nopember. Downloaded on March 29,2026 at 17:29:57 UTC from IEEE Xplore. Restrictions apply.

the determination of whether a DoS attack is occurring based on the reception of the ACK signal. Although it remains necessary to transmit a sampling packet to the controller initially to verify the ACK value, this approach establishes a basis for the subsequent design of the attack-related triggering condition and adaptive law.

D. AMETM With DoS Attacks

In order to describe the effects of energy-constrained DoS attacks on communication networks and system performance, we additionally introduce the following error term:

$$\tilde{e}(lkh) = x(lkh) - \tilde{x}(t_{k+1}h) \quad \forall l \in \mathbb{N} \tag{10}$$

where $lkh = t_k h + lh$, and $\tilde{x}(t_{k+1}h)$, as depicted in Fig. 2, represents the packet loss due to DoS attacks, but if there are no DoS attacks, it must have $\tilde{x}(t_{k+1}h) = x(t_{k+1}h)$.

From aforementioned analysis, the attack-related AMETM is formulated as follows:

$$t_{k+1}h = t_k h + \inf\left\{ lh \middle| \sum_{r=1}^{m} \lambda_r e_r^T(t)\Theta e_r(t) - \varrho(\mathsf{ACK})\tilde{e}^T(lkh)\Theta\tilde{e}(lkh) > \tilde{\varsigma}(t_k h)\bar{x}^T(t)\Theta\bar{x}(t) \right\} \tag{11}$$

where $\tilde{\varsigma}(t_k h)$ is the sampling value at $t = t_k h$ of the time-varying threshold function $\tilde{\varsigma}(t)$. This value can be regulated via the following attack-related adaptive law:

$$\dot{\tilde{\varsigma}}(t) = \left(\frac{1}{\tilde{\varsigma}^2(t)} - \frac{\kappa}{\tilde{\varsigma}(t)}\right)\sum_{r=1}^{m} \lambda_r e_r^T(t)\Theta e_r(t) - \left(\frac{1}{\tilde{\varsigma}^2(t)} - \frac{\tilde{\kappa}}{\tilde{\varsigma}(t)}\right)\varrho(\mathsf{ACK})\tilde{e}^T(lkh)\Theta\tilde{e}(lkh). \tag{12}$$

Here, $\tilde{\varsigma}(t) > 0$, scalars $\kappa > 0$ and $\tilde{\kappa} > 0$ are given to regulate the convergence rate of $\tilde{\varsigma}(t)$.

Motivated by [31] and [35], the effects of energy-constrained DoS attacks on ETMs can be considered as a bounded uncertainty item, that is

$$\tilde{e}^T(lkh)\Theta\tilde{e}(lkh) \le \overline{E} \tag{13}$$

where $\overline{E}$ represents the upper limit of uncertainty induced by energy-constrained DoS attacks.

Remark 5: Compared to the AMETM presented in (7), the attack-related AMETM (11) incorporates the ACK signal and packet loss error. This integration enables the determination of DoS attack occurrence by switching the ACK signal, while characterizing the ongoing packet loss caused by the attack through the bounded uncertainty item.

Remark 6: The attack-related triggering condition and adaptive law enable dynamic adjustment of the release probability in response to DoS attacks. During a DoS attack, when the ACK signal is lost ($\varrho(\mathsf{ACK}) = 1$), the triggering condition becomes more strict due to the attack-induced error term $\varrho(\mathsf{ACK})\tilde{e}^T(lkh)\Theta\tilde{e}(lkh)$, which reduces the release probability. This reduction decreases false-released sampling signals and conserves node energy. After the DoS attack ends and the event generator receives the ACK signal from the controller, $\varrho(\mathsf{ACK})$ switches from 1 to 0. Consequently, the attack-induced error term is deactivated, and both the triggering condition (11) and adaptive law (12) revert to the normal condition (7) and its corresponding adaptive law (8).

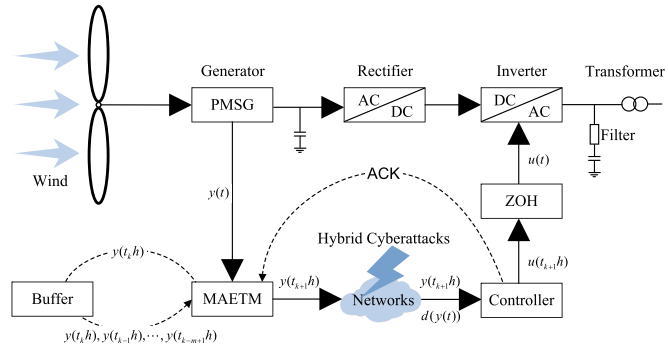

Fig. 3. Structure of PMSG-based WTS with AMETM under hybrid cyberattacks.

To facilitate analysis, the transmission delay of released signal $y(t_k h)$ is represented as $\tau_{t_k h}$, and the holding interval of the zero-order hold (ZOH) is denoted as $\Lambda = [t_k h + \tau_{t_k h}, t_{k+1}h + \tau_{t_{k+1}h})$, which can be divided into $\bigcup \Lambda_l$, where $\Lambda_l = [t_k h + lh + \mathcal{H}_{kl}, t_k h + (l+1)h + \mathcal{H}_{kl+1})$, $\forall l \in \mathbb{N}$. In addition, $\mathcal{H}_{kl} = \tau_{t_{k+1}h}$ if $l = t_{k+1} - t_k - 1$; otherwise, $\mathcal{H}_{kl} = \tau_{t_k h}$.

Define

$$\tau(t) = t - t_k h - lh, \ t \in \Lambda_l \tag{14}$$

where $0 \le \tau(t) \le \tau + h = \overline{\tau}$. The parameter τ represents the maximum communication delay $\tau_{t_k h}$. Then, the memory-based control $u(t)$ can be designed as

$$\begin{aligned} u(t) &= \sum_{r=1}^{m}\sum_{j=1}^{q} \theta_j(t_k h) K_{rj} y(t_{k-r+1}h) \\ &= \sum_{r=1}^{m}\sum_{j=1}^{q} \theta_j(t_k h) K_{rj} C(x(t-\tau(t)) - e_r(t)), t \in \Lambda_l. \end{aligned} \tag{15}$$

E. Random Deception Attacks

We assume that the adversaries have the ability to hijack the system output signal $y(t_k h)$ and tamper it as the malicious signal $d(y(t_k h))$. The structure of the WTS subjected to deception and DoS attacks is depicted in Fig. 3. In view of deception attacks occurring randomly, the controller is designed as follows:

$$\begin{aligned} u(t) = \beta(t)&\sum_{r=1}^{m}\sum_{j=1}^{q} \theta_j(t_k h) K_{rj} d_r(y(t_{k-r+1}h)) \\ &- (1-\beta(t))\sum_{r=1}^{m}\sum_{j=1}^{q} \theta_j(t_k h) K_{rj} y(t_{k-r+1}h), t \in \Lambda_l \end{aligned} \tag{16}$$

where $\beta(t) \in \{0, 1\}$ indicates the presence or no deception attack: $\beta(t) = 0$ signifies the absence of attack, while $\beta(t) = 1$ indicates that the packet sequence $\{y(t_k h), \ldots, y(t_{k-r+1}h)\}$

Authorized licensed use limited to: Institut Teknologi Sepuluh Nopember. Downloaded on March 29,2026 at 17:29:57 UTC from IEEE Xplore. Restrictions apply.

has been hijacked and altered with the malicious signals $\{d_1(y(t_k h)), \ldots, d_r(y(t_{k-r+1}h))\}$. The statistical properties of $\beta(t)$ are characterized as follows:

$$\mathbb{E}(\beta(t)) = \beta,\ \mathbb{E}((\beta(t) - \beta^2)) = \beta(1-\beta) \triangleq \beta_v^2 \tag{17}$$

where β represents the probability of deception attack occurrence. The malicious signals $d_r(Cx(t_{k-r+1}h))$, $r \in \{1, 2, \ldots, m\}$ are constrained to satisfy

$$||d_r(Cx(t_{k-r+1}h))||_2 \leq ||D_r Cx(t_{k-r+1}h)||_2 \tag{18}$$

where D_r are predefined constant matrices. For the sake of simplicity, let $D = \max\{D_r\}$.

Fig. 3 depicts the secure adaptive memory event-triggered control framework, where hybrid cyberattacks are considered exclusively in the sensor-to-controller channel. Evidently, the actuators will implement control actions with these successfully transmitted data in a ZOH style.

Remark 7: In communication networks, data is transmitted in packets, resulting in consistent resource consumption regardless of whether one or hundreds of bits are sent within a single payload. Consequently, if all historical release signals in AMETM (11) can be consolidated into one packet, the consumption of network resources will remain unchanged. Thus, m should be appropriately selected to satisfy the inequality $m \leq \aleph_p/\aleph_c - 1$, where $\aleph_p$ represents the payload of the packet in bits and $\aleph_c$ denotes the number of bits in a single release signal. For example, with $\aleph_p = 576$ bits (less than the 1472 bits permitted by the UDP protocol) and $\aleph_c = 8$ bits, it follows that $m \leq 71$, which is large enough for practical applications of the memory controller (16).

F. Closed-Loop System

Combing (6) and (16), the closed-loop system reads

$$\begin{aligned}\dot{x}(t) = &\sum_{i=1}^{q}\sum_{j=1}^{q}\theta_i\theta_j^k[A_i x(t) + F\varpi(t)\\ &+ \beta(t)\sum_{r=1}^{m} BK_{rj}d_r(y(t_{k-r+1}h))\\ &+ (1-\beta(t))\sum_{r=1}^{m} BK_{rj}C(x(t-\tau(t)) - e_r(t))], t \in \Lambda_l\end{aligned} \tag{19}$$

where θ_i and θ_j^k represent $\theta_i(t)$ and $\theta_j(t_k h)$, respectively. The initial condition for $x(t)$ is $x(t) = \chi(t)$, $t \in [-\overline{\tau}, 0]$.

Next, we introduce some necessary definitions for main results.

Definition 1 (see[36]): For a given function $V : C_{F_0}^b([-\overline{\tau}, 0], R^n) \times S$, its infinitesimal operator $\mathfrak{L}$ is defined as

$$\mathfrak{L}(V(t)) = \lim_{\Delta \to 0^+} \frac{1}{\Delta}[\mathbb{E}(V(t+\Delta)|t) - V(t)]. \tag{20}$$

Definition 2 (see[37]): If there exist positive constants p, q, $\forall r \in (0, q)$, and constant $\mathfrak{T}(r, p)$ such that $||x(t_0)|| < r$, and the following conclusion holds true:

$$||x(t)|| \leq p,\ \forall t > t_0 + \mathfrak{T}$$

then, the system is said to be uniformly ultimate bounded with the ultimate bound p.

Our aim is to design the controller (16) and attack-related AMETM (11) parameters to ensure that:

a) in the absence of DoS attacks and external disturbances, system (19) achieves mean-square asymptotic stability;
b) under the zero initial condition without DoS attacks, for any nonzero $\varpi(t) \in \mathcal{L}_2[0, \infty)$ and a given positive constant γ, system (19) meets the following H_∞ performance criterion: $||y(t)||_2 \leq \gamma||\varpi(t)||_2$;
c) in the presence of hybrid cyberattacks, system (19) maintains uniformly ultimate boundedness.

In addition, the performance loss is constrained to $||\mathcal{J}(x(lkh))|| \leq \mathcal{J}$, where $||\mathcal{J}(x(lkh))||$ denotes the performance loss resulting from hybrid cyberattacks, with $\mathcal{J}$ representing the maximum potential loss.

III. Main Results

This section is devoted to confirming that system (19) achieves $\mathcal{H}_\infty$ mean-square asymptotic stability when $\varrho(\mathsf{ACK}) = 0$, and to analyzing the security performance of the WTS under hybrid cyberattacks. Furthermore, sufficient conditions for the controller design are presented.

Theorem 1: For given positive constants h, τ, κ, $\tilde{\kappa}$, γ, β, β_v, λ_r and matrices K_{rj}, $r \in \{1, 2, \ldots, m\}$, $i, j \in \{1, 2, \ldots, q\}$, if there exist matrices $P > 0$, $R > 0$, $Q > 0$, $\Theta > 0$, and M satisfying

$$\sum_{i=1}^{q}\sum_{j=1}^{q}\theta_i\theta_j^k\Xi_{ij} < 0 \tag{21}$$

$$\begin{bmatrix} Q & M^T \\ * & Q \end{bmatrix} > 0 \tag{22}$$

then, system (19) achieves the following control objectives.

a) In the absence of DoS attacks, system (19) achieves mean-square asymptotic stability via AMETM (7) with the adaptive law (8) and meets H_∞ performance criterion $||y(t)||_2 \leq \gamma||\varpi(t)||_2$.
b) In the presence of hybrid cyberattacks, system (19) maintains uniformly ultimate boundedness via AMETM (11) with the adaptive law (12). At the same time, there is a constant $\mathfrak{T}$ that satisfies

$$\mathfrak{T} = \frac{\vartheta - \zeta}{-\lambda_{\max}(\Xi_{ij})\vartheta^2 - \tilde{\kappa}\overline{E}}$$

such that if the initial value satisfies

$$||x(t_0)|| > \zeta$$

then, for $\forall t \geq \mathfrak{T}$

$$||x(t)|| < \zeta$$

and the performance loss is expressed as

$$||\mathcal{J}(x(lkh))|| \leq \sqrt{\frac{\tilde{\kappa}\overline{E}}{\rho\lambda_{\min}(\mathcal{P})}}$$

Authorized licensed use limited to: Institut Teknologi Sepuluh Nopember. Downloaded on March 29,2026 at 17:29:57 UTC from IEEE Xplore. Restrictions apply.

where

$$\Xi_{ij} = \begin{bmatrix} \Sigma_1 & \Sigma_2 & \Sigma_4 \\ * & \Sigma_3 & 0 \\ * & * & \Sigma_5 \end{bmatrix} \Sigma_1 = \begin{bmatrix} \Sigma_{11} & \Sigma_{12} & \Sigma_{13} & \Sigma_{14} \\ * & \Sigma_{22} & 0 & 0 \\ * & * & \Sigma_{33} & 0 \\ * & * & * & \Sigma_{44} \end{bmatrix}$$

$$\Sigma_{11} = \begin{bmatrix} \Psi_{11} & \Psi_{12} & \Psi_{13} \\ * & \Psi_{22} & \Psi_{23} \\ * & * & \Psi_{33} \end{bmatrix} \Sigma_{12} = \begin{bmatrix} -(1-\beta)PN_{2j} \\ -\frac{\Theta}{m} \otimes I_{1\times m} \\ 0 \end{bmatrix}$$

$$N_{1j} = \sum_{r=1}^{m} BK_{rj}C \quad N_{2j} = B[K_{1j}C, \ldots, K_{mj}C]$$

$$\Psi_{11} = PA_i + A_i^T P + R - Q$$

$$\Psi_{12} = (1-\beta)PN_{1j} + Q - M^T$$

$$\Psi_{13} = M^T \quad \Psi_{22} = -2Q + M + M^T + \Theta$$

$$\Psi_{23} = Q - M^T \quad \Psi_{33} = -R - Q$$

$$\Sigma_{13} = \begin{bmatrix} \beta PN_{2j} \\ 0 \end{bmatrix} \quad \Sigma_{14} = \begin{bmatrix} PF \\ 0 \end{bmatrix}$$

$$\Sigma_{22} = \text{diag}\left\{\left(\frac{1}{m^2} - \kappa\lambda_1\right)\Theta, \ldots, \left(\frac{1}{m^2} - \kappa\lambda_m\right)\Theta\right\}$$

$$\Sigma_{33} = \text{diag}\{-I, \ldots, -I\} \quad \Sigma_{44} = -\gamma^2 I$$

$$\Sigma_2 = \begin{bmatrix} \overline{\tau}\mathcal{A}^T Q & \overline{\tau}\mathcal{B}^T Q \end{bmatrix} \quad \Sigma_3 = \text{diag}\{-Q, -Q\}$$

$$\Sigma_4 = \begin{bmatrix} C^T & \sqrt{m}C^T D^T \\ 0 & 0 \end{bmatrix} \quad \Sigma_5 = \text{diag}\{-I, -I\}$$

$$\mathcal{A} = \begin{bmatrix} A_i & (1-\beta)N_{1j} & 0 & -(1-\beta)N_{2j} & \beta N_{2j} & F \end{bmatrix}$$

$$\mathcal{B} = \begin{bmatrix} 0 & \beta_v N_{1j} & 0 & -\beta_v N_{2j} & -\beta_v N_{2j} & 0 \end{bmatrix}$$

$$\zeta = \sqrt{\frac{\tilde{\kappa}\overline{E}}{\lambda_{\max}(-\Xi_{ij})}} \quad \vartheta = \sqrt{\frac{V(0) + \frac{\tilde{\kappa}\overline{E}}{\rho}}{\lambda_{\min}(\mathcal{P})}}.$$

Proof: For system (19) under AMETM (11), the following Lyapunov functional candidate is selected:

$$V(t) = V_1(t) + V_2(t) + V_3(t) + V_4(t) \tag{23}$$

where

$$V_1(t) = x^T(t)Px(t)$$

$$V_2(t) = \int_{t-\overline{\tau}}^{t} x^T(s)Rx(s)ds$$

$$V_3(t) = \overline{\tau}\int_{t-\overline{\tau}}^{t}\int_{s}^{t} \dot{x}^T(\nu)Q\dot{x}(\nu)d\nu ds$$

$$V_4(t) = \frac{1}{2}\tilde{\varsigma}^2(t).$$

For any $t \in \Lambda_l$, $l \in \mathbb{N}$, applying the infinitesimal operator $\mathfrak{L}$ along the trajectories of system (19) and taking mathematical expectation on both sides yields

$$\begin{aligned}\mathbb{E}(\mathfrak{L}(V(t))) = {} & 2\sum_{i=1}^{q}\sum_{j=1}^{q}\theta_i\theta_j^k x^T(t)P\mathcal{A}\xi(t) \\ & + x^T(t)Rx(t) - x^T(t-\overline{\tau})Rx(t-\overline{\tau}) \\ & + \mathbb{E}\{\overline{\tau}^2\dot{x}^T(t)Q\dot{x}(t)\} - \overline{\tau}\int_{t-\overline{\tau}}^{t}\dot{x}^T(s)Q\dot{x}(s)ds \\ & + \tilde{\varsigma}(t)\dot{\tilde{\varsigma}}(t)\end{aligned} \tag{24}$$

where

$$\xi^T(t) = [x^T(t)x^T(t-\tau(t))x^T(t-\overline{\tau})E^T(t)D^T(y(t))\varpi^T(t)]$$

$$E^T(t) = [e_1^T(t), \ldots, e_m^T(t)]$$

$$D^T(y(t)) = [d_1^T(y(t)), \ldots, d_m^T(y(t))].$$

Combing (17) and (19) with straightforward calculations yields

$$\begin{aligned}& \mathbb{E}\{\overline{\tau}^2\dot{x}^T(t)Q\dot{x}(t)\} \\ & = \overline{\tau}^2\sum_{i=1}^{q}\sum_{j=1}^{q}\theta_i\theta_j^k\xi^T(t)(\mathcal{A}^T Q\mathcal{A} + \mathcal{B}^T Q\mathcal{B})\xi(t).\end{aligned} \tag{25}$$

From (22) and using Park's theorem in [38], one has

$$\begin{aligned}& -\overline{\tau}\int_{t-\overline{\tau}}^{t}\dot{x}^T(s)Q\dot{x}(s)ds \\ \leq & -[x(t-\tau(t)) - x(t-\overline{\tau})]^T Q[x(t-\tau(t)) - x(t-\overline{\tau})] \\ & -[x(t) - x(t-\tau(t))]^T Q[x(t) - x(t-\tau(t))] \\ & -2[x(t-\tau(t)) - x(t-\overline{\tau})]^T M[x(t) - x(t-\tau(t))].\end{aligned} \tag{26}$$

Recalling AMETM (7) and adaptive law (8), one obtains

$$\begin{aligned}\tilde{\varsigma}(t)\dot{\tilde{\varsigma}}(t) = & -\kappa\sum_{r=1}^{m}\lambda_r e_r^T(t)\Theta e_r(t) \\ & + \frac{1}{\tilde{\varsigma}(t)}\sum_{r=1}^{m}\lambda_r e_r^T(t)\Theta e_r(t) \\ \leq & \bar{x}^T(t)\Theta\bar{x}(t) - \kappa\sum_{r=1}^{m}\lambda_r e_r^T(t)\Theta e_r(t).\end{aligned} \tag{27}$$

In addition, recalling the constraint in (18), one has

$$\begin{aligned}& \sum_{r=1}^{m}\left[(x^T(t-\tau(t)) - e_r(t))C^T D^T DC(x(t-\tau(t)) - e_r(t))\right. \\ & \left. -d_r^T(y(t))d_r(y(t))\right] \geq 0.\end{aligned} \tag{28}$$

Utilizing the Schur complement, one can conclude from (21), (22) and (24)–(28) that

$$\mathbb{E}(\mathfrak{L}(V(t))) \leq -y^T(t)y(t) + \gamma^2\varpi^T(t)\varpi(t). \tag{29}$$

Since $\mathbb{E}(V(t))$ is continuous in t, it follows from (29) that

$$\mathbb{E}(V(\infty)) - V(0) \leq \int_0^{\infty}\gamma^2\varpi^T(t)\varpi(t) - y^T(t)y(t)dt.$$

Authorized licensed use limited to: Institut Teknologi Sepuluh Nopember. Downloaded on March 29,2026 at 17:29:57 UTC from IEEE Xplore. Restrictions apply.

Under the zero initial condition, the following inequality can be established:

$$\int_0^\infty y^T(t)y(t)dt \le \int_0^\infty \gamma\varpi^T(t)\varpi(t)dt,$$

that is $||y(t)||_2 \le \gamma||\varpi(t)||_2$. Therefore, system (19) achieves mean-square asymptotic stability and possesses H_∞ performance in the absence of DoS attacks.

Next, consider the security performance of system (19) under hybrid cyberattacks. By considering the attack-related triggering condition (11) and adaptive law (12), one has

$$\mathbb{E}(\mathfrak{L}(V(t))) \le \sum_{i=1}^{q}\sum_{j=1}^{q}\theta_i\theta_j^k\xi^T(t)\Xi_{ij}\xi(t) + \tilde{\kappa}\tilde{e}^T(lkh)\Theta\tilde{e}(lkh).$$

Recalling $0 \le \tilde{e}^T(lkh)\Theta\tilde{e}(lkh) \le \overline{E}$ in (13), it follows that

$$\mathbb{E}(\mathfrak{L}(V(t))) \le \sum_{i=1}^{q}\sum_{j=1}^{q}\theta_i\theta_j^k\xi^T(t)\Xi_{ij}\xi(t) + \tilde{\kappa}\overline{E}. \tag{30}$$

Considering (30), $\mathbb{E}(\mathfrak{L}(V(t))) < 0$ only if

$$||\xi(t)|| > \zeta \tag{31}$$

where $\zeta = [\tilde{\kappa}\overline{E}/(\lambda_{\max}(-\Xi_{ij}))]^{\frac{1}{2}}$. Therefore, $\mathbb{E}(\mathfrak{L}(V(t)))$ exhibits negative definiteness outside the set $\Gamma_\zeta \triangleq \{\xi|||\xi(t)|| \le \zeta\}$, which indicates that $V(t)$ decreases as $||\xi(t)||$ departs from the set Γ_ζ, and $||\xi(t)||$ is constrained within the ultimate bound ζ. Moreover, since $\sum_{i=1}^{q}\sum_{j=1}^{q}\theta_i\theta_j^k\Xi_{ij} < 0$, there exist a suitable parameter $\rho > 0$ such that

$$\mathbb{E}(\mathfrak{L}(V(t))) \le -\rho V(t) + \tilde{\kappa}\overline{E}. \tag{32}$$

Multiplying both sides of (32) by $e^{-\rho t}$ and integrating them, one has

$$\begin{aligned}\mathbb{E}(V(t)) &\le e^{-\rho t}V(0) + \frac{\tilde{\kappa}\overline{E}}{\rho}(1-e^{-\rho t})\\ &\le V(0) + \frac{\tilde{\kappa}\overline{E}}{\rho}.\end{aligned}$$

In addition, from (23), it is evident that

$$\xi^T(t)\mathcal{P}\xi(t) \le \mathbb{E}(V(t)) \le V(0) + \frac{\tilde{\kappa}\overline{E}}{\rho} \tag{33}$$

where $\mathcal{P} = \mathrm{diag}\{P, 0, \ldots, 0\}$. Thus

$$\zeta < ||\xi(t)|| \le \vartheta = \sqrt{\frac{V(0) + \frac{\tilde{\kappa}\overline{E}}{\rho}}{\lambda_{\min}(\mathcal{P})}} \tag{34}$$

and the performance loss can be expressed as

$$||\mathcal{J}(x(lkh))|| = \sqrt{\frac{\tilde{\kappa}\tilde{e}^T(lkh)\Theta\tilde{e}(lkh)}{\rho\lambda_{\min}(\mathcal{P})}} \le \sqrt{\frac{\tilde{\kappa}\overline{E}}{\rho\lambda_{\min}(\mathcal{P})}} = \mathcal{J}.$$

Finally, consider the time $\mathcal{T}$ required for the state trajectories to converge to the set $\Gamma_\zeta^\vartheta \triangleq \{\xi|\zeta < \mathbb{E}\{||\xi(t)||\} \le \vartheta\}$. From (21) and (31), there exists a positive constant σ satisfying

$$\sigma = -\min_{\xi(t)\in\Gamma_\vartheta}\sum_{i=1}^{q}\sum_{j=1}^{q}\theta_i\theta_j^k\xi^T(t)\Xi_{ij}\xi(t) - \tilde{\kappa}\overline{E}.$$

It then follows that

$$\sigma = -\lambda_{\max}(\Xi_{ij})\vartheta^2 - \tilde{\kappa}\overline{E}.$$

Hence

$$V(t) \le V(t_0) - (t-t_0)\sigma \le \vartheta - (t-t_0)\sigma$$

which means that $V(t)$ will decrease to ζ within $[t_0, t_0 + \frac{\vartheta-\zeta}{\sigma}]$, that is

$$\mathcal{T} = \frac{\vartheta-\zeta}{\sigma} = \frac{\vartheta-\zeta}{-\lambda_{\max}(\Xi_{ij})\vartheta^2 - \tilde{\kappa}\overline{E}}.$$

Consequently, given any initial condition such that $||\xi(t)|| < \vartheta$, the state variables will converge to the secure set Γ_ζ^ϑ after the finite time $\mathcal{T}$, i.e., $||\xi(t)|| \le \zeta$. Since $||x(t)|| \le ||\xi(t)||$, one can get

$$||x(t)|| \le \zeta.$$

From the analysis above, we can deduce that there exists a time constant $\mathcal{T}$, which ensures that all state trajectories originating in the ball Γ_ϑ will converge to the smaller ball Γ_ζ^ϑ with the ultimate bound ζ. □

Considering the complexities of asynchronous premise variables and mismatched membership functions, Theorem 2 employs the characteristics of membership functions and slack matrices Λ_i, $i \in \{1, 2, \ldots, q\}$, to achieve more relaxed results.

Theorem 2: For given positive constants h, τ, κ, $\tilde{\kappa}$, γ, β, β_v, λ_r and matrices K_{rj}, $r \in \{1, 2, \ldots, m\}$, $i, j \in \{1, 2, \ldots, q\}$, if the membership functions satisfy $\theta_i - \delta_j\theta_j^k > 0$ $(0 < \delta_j \le 1)$, and there exist matrices $P > 0$, $R > 0$, $Q > 0$, $\Theta > 0$, and M, Λ_i satisfying

$$\Xi_{ij} - \Lambda_i < 0 \tag{35}$$

$$\delta_i\Xi_{ii} - \delta_i\Lambda_i + \Lambda_i < 0 \tag{36}$$

$$\delta_j(\Xi_{ij} - \Lambda_i) + \Lambda_i + \delta_i(\Xi_{ji} - \Lambda_j) + \Lambda_j < 0 \ (i < j) \tag{37}$$

$$\begin{bmatrix} Q & M^T \\ * & Q \end{bmatrix} \ge 0 \tag{38}$$

then, system (19) achieves the control objective (i) and objective (ii) in Theorem 1.

Proof: By utilizing the assumption $\theta_j^k - \delta_j\theta_j > 0$ in [26] and slack matrices Λ_i, the mismatched membership functions can be addressed to yield

$$\begin{aligned}&\sum_{i=1}^{q}\sum_{j=1}^{q}\theta_i\theta_j^k\Xi_{ij}\\ &= \sum_{i=1}^{q}\theta_i^2(\delta_i\Xi_{ii} - \delta_i\Lambda_i + \Lambda_i) + \sum_{j=1}^{q}\sum_{i<j}^{q}\theta_i\theta_j\\ &\quad\times(\delta_j\Xi_{ij} - \delta_j\Lambda_i + \Lambda_i + \delta_i\Xi_{ji} - \delta_i\Lambda_j + \Lambda_j)\\ &\quad+ \sum_{i=1}^{q}\sum_{j=1}^{q}\theta_i(\theta_j^k - \delta_j\theta_j)(\Xi_{ij} - \Lambda_i).\end{aligned} \tag{39}$$

Authorized licensed use limited to: Institut Teknologi Sepuluh Nopember. Downloaded on March 29,2026 at 17:29:57 UTC from IEEE Xplore. Restrictions apply.

Algorithm 1: Calculate Control Input $u(t)$.

Input: h, τ, κ, $\tilde{\kappa}$, γ, β, β_v, λ_r, δ_j, A_i, B, C, F_i;
Output: $u(t)$;
if *Theorem 3 yields no feasible solution* **then**
  return to **Input** and adjust given constants;
else
  $K_{rj} = \tilde{K}_{rj} X^{-1} C^T (CC^T)^{-1}$, $\Theta = X^{-1}\tilde{\Theta}X^{-1}$;
end
Set the simulation time T and generate an energy-constrained DoS attack sequence.
Define $\hat{\beta}$ to represent the probability of a deception attack actual occurrence.
for $t = 0 : h : T$ **do**
  if $t \notin \mathbb{H}(0, T)$ **then**
    $\varrho(\mathsf{ACK}) = 0$, $\hat{\beta} = \beta$, using AMETM (7);
    if *triggering condition (7) is satisfied* **then**
      $t_{k-m+1}h = t_{k-m}h$,
      $\vdots$
      $t_k h = t_{kh} + lh$;
    end
    if *deception attack occurs* **then**
      $u(t) = \sum_{r=1}^{m} Kd(t_{k-r+1}h)$;
    else
      $u(t) = \sum_{r=1}^{m} Ky(t_{k-r+1}h)$;
    end
  else
    $\varrho(\mathsf{ACK}) = 1$, using attack-related AMETM (11);
    if *the last transmitted data packet is tampered with by the deception attack* **then**
      $u(t) = \sum_{r=1}^{m} Kd(t_{k-r}h)$, $\hat{\beta} = 1$;
    else
      $u(t) = \sum_{r=1}^{m} Ky(t_{k-r}h)$, $\hat{\beta} = 0$;
    end
  end
end

Based on (35)–(37), one has $\sum_{i=1}^{q}\sum_{j=1}^{q}\theta_i\theta_j^k\Xi_{ij} < 0$, which indicates that system (19) achieves control objectives (i) and (ii) in Theorem 1. □

Theorem 3: For given positive constants h, τ, κ, $\tilde{\kappa}$, γ, β, β_v, λ_r, $r \in \{1, 2, \ldots, m\}$, if the membership functions satisfy $\theta_i - \delta_j\theta_j^k > 0$ $(0 < \delta_j \le 1)$, $i, j \in \{1, 2, \ldots, q\}$, and there exist matrices $X > 0$, $\tilde{R} > 0$, $\tilde{Q} > 0$, $\tilde{\Theta} > 0$, and $\tilde{M}$, $\tilde{\Lambda}_i$, $\tilde{K}_{rj}$ satisfying

$$\tilde{\Xi}_{ij} - \tilde{\Lambda}_i < 0 \tag{40}$$

$$\delta_i\tilde{\Xi}_{ii} - \delta_i\tilde{\Lambda}_i + \tilde{\Lambda}_i < 0 \tag{41}$$

$$\delta_j(\tilde{\Xi}_{ij} - \tilde{\Lambda}_i) + \tilde{\Lambda}_i + \delta_i(\tilde{\Xi}_{ji} - \tilde{\Lambda}_j) + \tilde{\Lambda}_j < 0 \ (i < j) \tag{42}$$

$$\begin{bmatrix} \tilde{Q} & \tilde{M}^T \\ * & \tilde{Q} \end{bmatrix} \ge 0 \tag{43}$$

where

$$\tilde{\Xi}_{ij} = \begin{bmatrix} \tilde{\Sigma}_1 & \tilde{\Sigma}_2 & \tilde{\Sigma}_4 \\ * & \tilde{\Sigma}_3 & 0 \\ * & * & \tilde{\Sigma}_5 \end{bmatrix} \tilde{\Sigma}_1 = \begin{bmatrix} \tilde{\Sigma}_{11} & \tilde{\Sigma}_{12} & \tilde{\Sigma}_{13} & \tilde{\Sigma}_{14} \\ * & \tilde{\Sigma}_{22} & 0 & 0 \\ * & * & \tilde{\Sigma}_{33} & 0 \\ * & * & * & \tilde{\Sigma}_{44} \end{bmatrix}$$

$$\tilde{\Sigma}_{11} = \begin{bmatrix} \tilde{\Sigma}_{11} & \tilde{\Sigma}_{12} & \tilde{\Sigma}_{13} \\ * & \tilde{\Sigma}_{22} & \tilde{\Sigma}_{23} \\ * & * & \tilde{\Sigma}_{33} \end{bmatrix} \tilde{\Sigma}_{12} = \begin{bmatrix} -(1-\beta)\tilde{N}_{2j} \\ -\frac{\tilde{\Theta}}{m} \otimes I_{1\times m} \\ 0 \end{bmatrix}$$

$$\tilde{N}_{1j} = \sum_{r=1}^{m} B\tilde{K}_{rj} \quad \tilde{N}_{2j} = B[\tilde{K}_{1j}, \ldots, \tilde{K}_{mj}]$$

$$\tilde{\Psi}_{11} = A_i X + XA_i^T + \tilde{R} - \tilde{Q}$$

$$\tilde{\Psi}_{12} = (1-\beta)\tilde{N}_{1j} + \tilde{Q} - \tilde{M}^T$$

$$\tilde{\Psi}_{13} = \tilde{M}^T \quad \tilde{\Psi}_{22} = -2\tilde{Q} + \tilde{M} + \tilde{M}^T + \tilde{\Theta}$$

$$\tilde{\Psi}_{23} = \tilde{Q} - \tilde{M}^T \quad \tilde{\Psi}_{33} = -\tilde{R} - \tilde{Q}$$

$$\tilde{\Sigma}_{13} = \begin{bmatrix} \beta\tilde{N}_{2j} \\ 0 \end{bmatrix} \quad \Sigma_{14} = \begin{bmatrix} F \\ 0 \end{bmatrix}$$

$$\tilde{\Sigma}_{22} = \mathrm{diag}\left\{\left(\frac{1}{m^2} - \kappa\lambda_1\right)\tilde{\Theta}, \ldots, \left(\frac{1}{m^2} - \kappa\lambda_m\right)\tilde{\Theta}\right\}$$

$$\tilde{\Sigma}_{33} = \mathrm{diag}\{-I, \ldots, -I\} \quad \tilde{\Sigma}_{44} = -\gamma^2 I$$

$$\tilde{\Sigma}_2 = \begin{bmatrix} \overline{\tau}\tilde{\mathcal{A}}^T & \overline{\tau}\tilde{\mathcal{B}}^T \end{bmatrix} \quad \tilde{\Sigma}_3 = \mathrm{diag}\left\{\tilde{Q} - 2X, \tilde{Q} - 2X\right\}$$

$$\tilde{\Sigma}_4 = \begin{bmatrix} C^T G^T & \sqrt{m}C^T G^T D^T \\ 0 & 0 \end{bmatrix} \quad \tilde{\Sigma}_5 = \mathrm{diag}\{-I, -I\}$$

$$\tilde{\mathcal{A}} = \begin{bmatrix} A_i X & (1-\beta)\tilde{N}_{1j} & 0 & -(1-\beta)\tilde{N}_{2j} & \beta\tilde{N}_{2j} & F \end{bmatrix}$$

$$\tilde{\mathcal{B}} = \begin{bmatrix} 0 & \beta_v\tilde{N}_{1j} & 0 & -\beta_v\tilde{N}_{2j} & -\beta_v\tilde{N}_{2j} & 0 \end{bmatrix}$$

then, system (19) achieves the control objectives (i) and (ii) in Theorem 1, and the controller gains are computed as $K_{rj} = \tilde{K}_{rj}X^{-1}C^T(CC^T)^{-1}$.

Proof: Define $X = P^{-1}$, $\tilde{R} = XRX$, $\tilde{Q} = XQX$, $\tilde{\Theta} = X\Theta X$, $\tilde{M} = XMX$, and $\tilde{\Lambda}_i = X\Lambda_i X$, left multiply and right multiply both sides of (35)–(37) by $\mathrm{diag}\{\underbrace{X, X, X, \ldots, X}_{3+2m}, I, Q^{-1}, Q^{-1}, I, I\}$, and replace $K_{rj}CX$ with $\tilde{K}_{rj}$. Since CC^T in (19) is full row rank, $K_{rj} = \tilde{K}_{rj}X^{-1}C^T(CC^T)^{-1}$. Furthermore, the nonlinear item $X\tilde{Q}^{-1}X$ can be replaced by $\tilde{Q} - 2X$ because of $(X - \tilde{Q})\tilde{Q}^{-1}(X - \tilde{Q}) > 0$. Based on the above analysis, (40)–(42) can be obtained from (35) to (37). In addition, left and right multiply $\mathrm{diag}\{X, X\}$ for (37), inequality (43) can be obtained as well.

Moreover, considering the influence of hybrid cyberattacks and the design of control gains K_{rj}, Algorithm 1 is proposed based on Theorem 3 to calculate the control input $u(t)$. □

Authorized licensed use limited to: Institut Teknologi Sepuluh Nopember. Downloaded on March 29,2026 at 17:29:57 UTC from IEEE Xplore. Restrictions apply.

IV. Numerical Simulations

This section validates the effectiveness and advantages of the proposed AMETM-based control approach by using a WTS with surface-mounted PMSG as an example. The physical parameters are provided in Table I.

Suppose that the deception attack $d_r(y(t)) = -\tanh(0.2y(t))$. It follows that $||d_r(y(t))|| \leq ||Dy(t)||_2$, where $D = 0.2$. Other parameters are specified as $h = \tau = 0.001s$, $\kappa = 12$, $\tilde{\kappa} = 15$, $\beta = 0.3$, $\beta_v = 0.2$, $\gamma = 5$, $\delta_1 = \delta_2 = 0.9$, $\mathbb{D}_1 = -50.35$, $\mathbb{D}_2 = 50.35$, and the initial value is set to be $x(0) = [0.5\ 0.1\ -0.007]^T$.

Two distinct scenarios are examined for comparison. In Scenario 1, it is assumed that only deception attacks occur. Under this circumstance, the triggering mechanism simplifies to AMETM (7). Compared to the memory event-triggered mechanism (METM) [24], which utilizes a fixed threshold, the memory-based adaptive law in AMETM (7) dynamically adjusts the triggering threshold to enhance communication efficiency.

Scenario 2 assumes hybrid cyberattacks occur. The simulation results illustrate that, in contrast to the AMETM (7), the proposed attack-related AMETM (11) eliminates false-triggered events significantly during DoS attacks. Moreover, in comparison to the memoryless attack-related AETM [31], the introduced historical release signals speed up the communication frequency during state fluctuations, leading to a smoother system response.

Scenario 1: Consider the scenario where only deception attacks occur, with the external disturbance defined as:

$$\begin{cases} \varpi(t) = -\sin(t), & t \in [0, 10] \\ \varpi(t) = 0, & t \in (10, \infty). \end{cases} \tag{44}$$

First, by solving inequalities in Theorem 3 with a buffer size of $m = 3$, and selecting the weighted parameters as $\lambda_1 = 0.5$, $\lambda_2 = 0.3$, $\lambda_3 = 0.2$, the feedback gains are determined as

$$K_{11} = -[7.8033\ 7.8056]^T,\ K_{12} = -[7.7751\ 7.8154]^T$$

$$K_{21} = -[4.5066\ 4.4932]^T,\ K_{22} = -[4.4855\ 4.4987]^T$$

$$K_{31} = -[4.9327\ 4.9164]^T,\ K_{32} = -[4.9156\ 4.9155]^T.$$

The simulation results are illustrated in Figs. 4–7 for $t \in [0, 35]$. Fig. 4 presents the chaotic behavior of system (19) without control, which demonstrates the unstable dynamics of the open-loop system. Fig. 5 plots the response of the closed-loop T–S fuzzy system (19). Despite the interference from deception attacks and external disturbances, the state variables still converge quickly to zero. Figs. 6 and 7 depict respectively the release instants and intervals of AMETM (7), along with the trajectory of threshold function $\varsigma(t)$. It can be seen that in the early stage of control, the system state fluctuates significantly. To speed up state convergence, the memory-based adaptive law (8) lowers the triggering threshold, which increases the communication frequency and improves the control effect.

Next, to investigate the influence of different weighted parameters λ_r, four different groups of λ_r, $r \in \{1, 2, 3\}$, are selected to illustrate the relationship among the weighted parameter λ_r, the number of released packets, and the norm of the output response

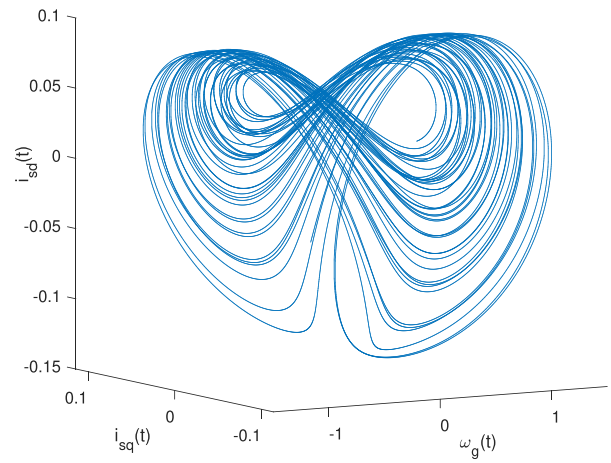

Fig. 4. Phase portraits of $\omega_g(t)$, $\mathbb{i}_{sq}(t)$, and $\mathbb{i}_{sd}(t)$ without control inputs.

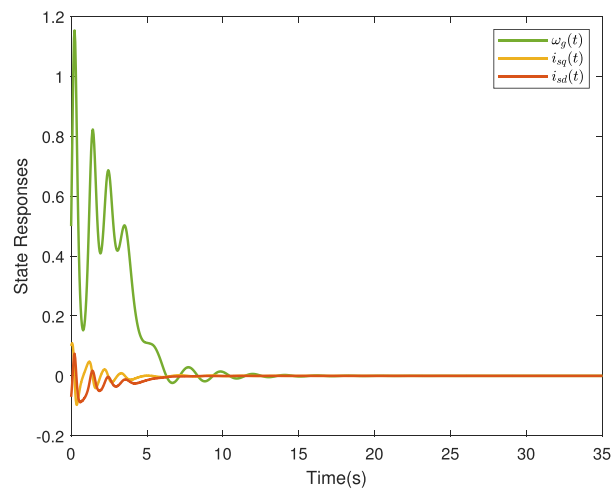

Fig. 5. State responses $\omega_g(t)$, $\mathbb{i}_{sq}(t)$, and $\mathbb{i}_{sd}(t)$ with control inputs $\mathbb{V}_{sq}(t)$ and $\mathbb{V}_{sd}(t)$ under external disturbance and deception attacks.

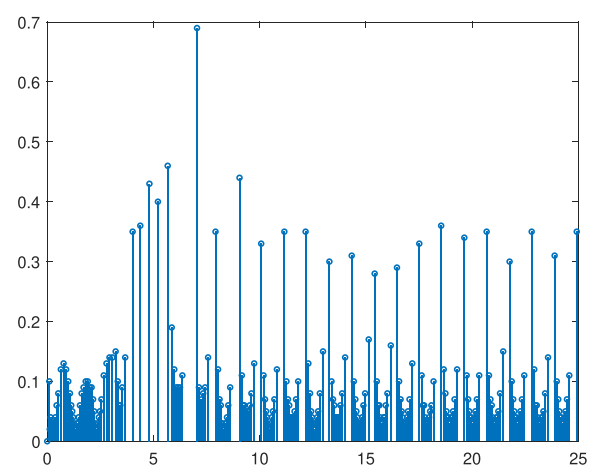

Fig. 6. Release intervals under external disturbance and deception attacks.

Authorized licensed use limited to: Institut Teknologi Sepuluh Nopember. Downloaded on March 29,2026 at 17:29:57 UTC from IEEE Xplore. Restrictions apply.

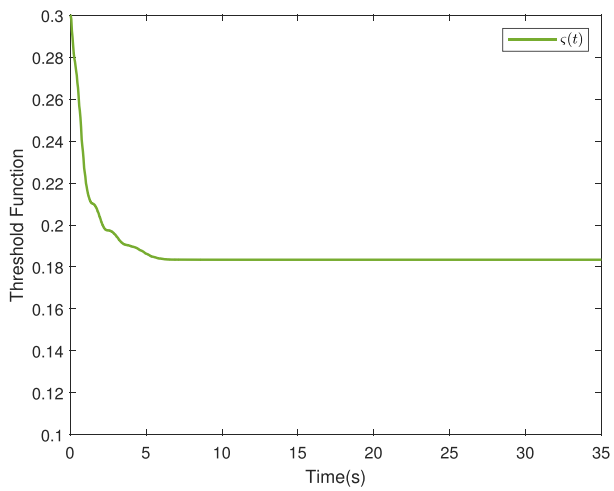

Fig. 7. Threshold function $\varsigma(t)$ under external disturbance and deception attacks.

TABLE II
COMPARISON OF THE NUMBER OF RELEASED DATA PACKETS AND OUTPUT RESPONSE NORM FOR DIFFERENT GROUPS OF λ_r ($r \in \{1, 2, 3\}$)

Groups	λ_1	λ_2	λ_3	SRP	$\|\|y(t)\|\|_2$
1	1/3	1/3	1/3	634	2359
2	0.5	0.3	0.2	543	2438
3	0.7	0.2	0.1	438	2464
4	0.9	0.05	0.05	333	2566

$||y(t)||_2$, as shown in Table II. For simplicity, SRP and URP are utilized to denote successfully and unsuccessfully released packets, respectively.

Table II demonstrates that variations in λ_r significantly influence both the number of released packets and the output response. Notably, the effect of λ_1 is the most pronounced, indicating that recent data transmissions are more important than historical ones. In other words, $\lambda_r \geq \lambda_{r+1}$. Furthermore, the increase in λ_1 correlates with the decrease in communication frequency and system performance. Consequently, it is essential to make a tradeoff between communication resources and control effects based on the actual situation.

Scenario 2: Consider the scenario where hybrid cyberattacks occur. The disturbance is selected identically to that in scenario 1. First, by using the attack-related AMETM (11) and applying Theorem 3 with $m = 3$, $\lambda_1 = 0.7$, $\lambda_2 = 0.2$, and $\lambda_3 = 0.1$, the following feedback gains are derived:

$$K_{11} = -[10.3829\ 10.3579]^T,\ K_{12} = -[10.3416\ 10.3709]^T$$

$$K_{21} = -[3.8495\ 3.8248]^T,\ K_{22} = -[3.8315\ 3.8325]^T$$

$$K_{31} = -[2.7856\ 2.7991]^T,\ K_{32} = -[2.7843\ 2.8083]^T.$$

Simulation results are shown in Fig. 8 for $t \in [0, 35]$, which includes the following:

a) the schematic diagram of hybrid cyberattacks;
b) state responses;

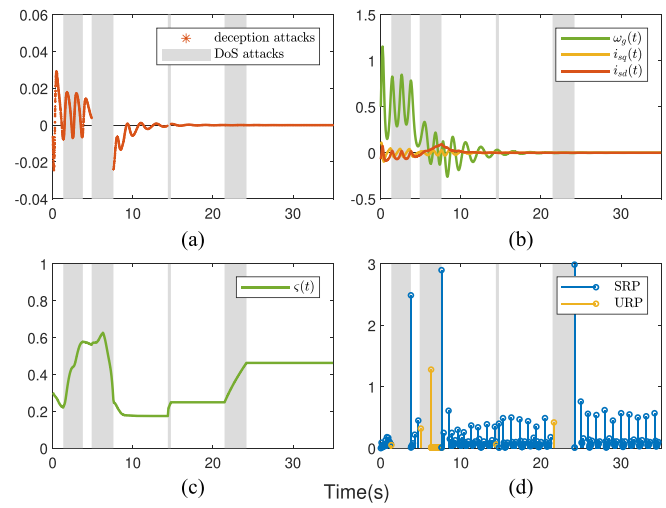

Fig. 8. Simulation results for PMSG-based WTS under hybrid cyberattacks by using attack-related AMETM (11). (a) Hybrid cyberattacks. (b) State responses. (c) Threshold function trajectory. (d) Successfully/unsuccessfully release instants and intervals.

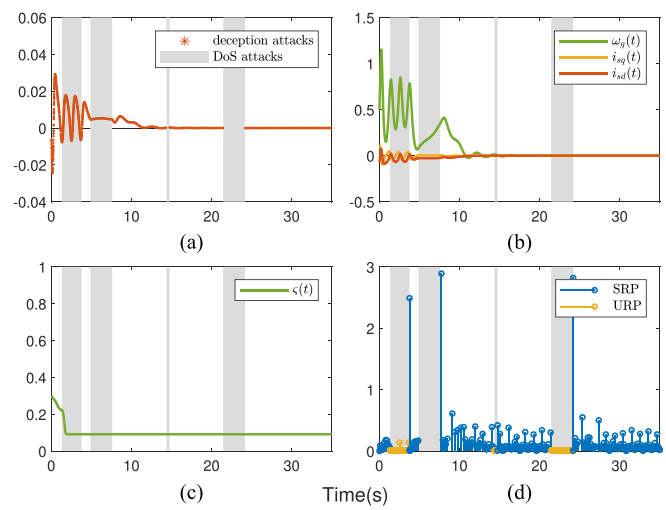

Fig. 9. Simulation results for PMSG-based WTS under hybrid cyberattacks by using AMETM (7). (a) Hybrid cyberattacks. (b) State responses. (c) Threshold function trajectory. (d) Successfully/unsuccessfully release instants and intervals.

c) the threshold function trajectory;
d) successfully/unsuccessfully release instants and intervals.

To validate the efficacy of the proposed attack-related AMETM (11) in defending against hybrid cyberattacks and enhancing communication efficiency, two sets of simulation results are provided in Figs. 9 and 10 for comparison. These results are based on AMETM (7) and memoryless attack-related AETM [31], respectively. The images in these simulation results correspond to those in Fig. 8.

Figs. 8–10(a) depict energy-constrained DoS attacks and randomly occurring deception attacks. It is worth noting that hybrid cyberattacks can cause substantial damage to the system compared to single deception attacks. When DoS attacks occur, if the last transmitted data packet is tampered with by a deception

Authorized licensed use limited to: Institut Teknologi Sepuluh Nopember. Downloaded on March 29,2026 at 17:29:57 UTC from IEEE Xplore. Restrictions apply.

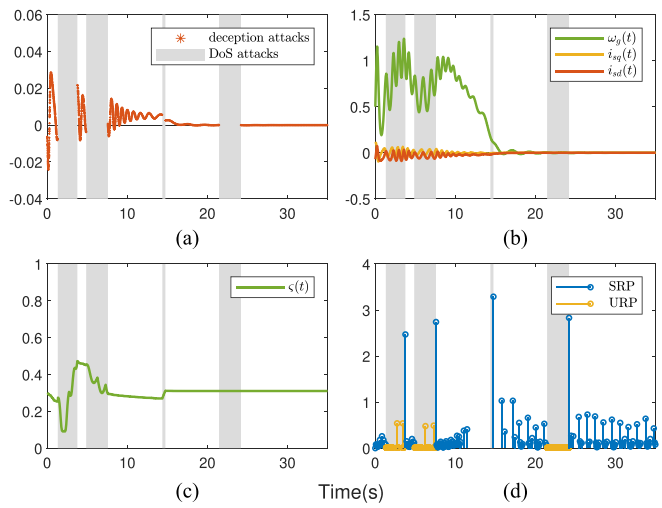

Fig. 10. Simulation results for PMSG-based WTS under hybrid cyberattacks by using memoryless attack-related AETM [31]. (a) Hybrid cyberattacks. (b) State responses. (c) Threshold function trajectory. (d) Successfully/unsuccessfully release instants and intervals.

TABLE III
COMPARISON OF THE NUMBER OF RELEASED PACKETS AND OUTPUT RESPONSE NORM BASED ON DIFFERENT ETMS

	SRP	URP	$\|\|y(t)\|\|_2$
attack-related AMETM (11)	261	134	3427
AMETM (7)	381	531	3689
attack-related AETM [31]	163	564	10513

attack, the controller will continuously utilize the tampered control signal under the action of the ZOH until the end of this DoS attack period. This phenomenon is exemplified in Algorithm 1 and Figs. 8(a) and 9(a), where a persistent deception signal exists throughout the DoS attack duration.

Figs. 8–10(b) plot the state trajectory under hybrid cyberattacks, revealing a more pronounced performance loss compared to the scenario in Fig. 5, which is subject to singular deception attacks. Notably, the state variables in Fig. 8(b) exhibit a quicker convergence than those in Figs. 9(b) and 10(b). This difference can be quantitatively assessed by comparing the output response norm $||y(t)||_2$ presented in Table III. Specifically, the $||y(t)||_2$ under the attack-related AMETM is 7.1% smaller than that under the AMETM (7) and 67.4% smaller than that under the attack-related AETM [31]. These results reflect the ETM proposed in this article exhibits strong resistance against hybrid cyberattacks.

Figs. 8–10(c) and (d) present the trajectory of the threshold function $\varsigma(t)$, as well as the release instants and intervals under the adaptive laws. It is evident that, in contrast to the adaptive law depicted in Fig. 9(c), the attack-related adaptive laws in Figs. 8(c) and 10(c) can adjust dynamically the triggering threshold based on the occurrence of DoS attacks. As shown in Figs. 8(d)–10(d) and Table III, the total number of released packets (including SRP and URP) under the attack-related AMETM (11) decreased significantly by 56.69% and 45.67%, respectively, compared to the other two ETMs (AMETM (7) and attack-related AETM [31]). The reduction in URP was even more pronounced, at 74.76% and 76.24%, respectively. This demonstrates that the attack-related triggering condition and adaptive law in this study reduces significantly unnecessary data transmission during cyberattacks. Furthermore, while attack-related conditions were also introduced in AETM [31], the number of URP did not decrease; instead,the SRP number decreased. This discrepancy may stem from the neglect of crucial sampling data packets during intense state fluctuations in memoryless ETMs, leading to poorer control efficacy compared to the other two ETMs and limiting the effectiveness of the introduced attack-related terms.

V. CONCLUSION

This article presented a novel attack-related AMETM for WTS under hybrid cyberattacks. First, a unified model of the T–S fuzzy PMSG-based WTS was established, which integrated event-triggered communication, network-induced delays, asynchronous premise variables, and hybrid cyberattacks. Second, the weaknesses of conventional ETMs in countering attacks and conserving communication resources were discussed. An attack-related AMETM and adaptive law were designed to resist cyberattacks and improve communication efficiency. Third, sufficient conditions were provided to ensure the mean-square $\mathcal{H}_\infty$ asymptotic stability of the resulting fuzzy system in the absence of DoS attacks, while guaranteeing the uniformly ultimate boundedness in their presence. Finally, the effectiveness of the proposed approach was validated through numerical simulations. Future research will explore broader and more intricate scenarios, specifically investigating the implementation of the attack-related AMETM in contexts involving unbounded deception attacks.

REFERENCES

[1] D. Mukherjee and A. Banerjee, "DC grid interface for the integrated generator–rectifier architecture in wind energy systems," *IEEE Trans. Power Electron.*, vol. 37, no. 12, pp. 14795–14808, Dec. 2022.
[2] G. S. Chawda, A. G. Shaik, O. P. Mahela, and S. Padmanaban, "Performance improvement of weak grid-connected wind energy system using FLSRF-controlled DSTATCOM," *IEEE Trans. Ind. Electron.*, vol. 70, no. 2, pp. 1565–1575, Feb. 2023.
[3] S. Rasool, K. M. Muttaqi, and D. Sutanto, "A novel configuration of a hybrid offshore wind-wave energy conversion system and its controls for a remote area power supply," *IEEE Trans. Ind. Appl.*, vol. 58, no. 6, pp. 7805–7817, Nov./Dec. 2022.
[4] A. A. Basheer, K. Palanimuthu, S. R. Lee, and Y. H. Joo, "Efficiency enhancement using fault-tolerant sliding mode control for PMVG-based WTS under actuator faults," *IEEE Trans. Ind. Electron.*, vol. 71, no. 1, pp. 513–523, Jan. 2024.
[5] K. Palanimuthu, G. Mayilsamy, S. R. Lee, S. Y. Jung, and Y. H. Joo, "Fault ride-through for PMVG-based wind turbine system using coordinated active and reactive power control strategy," *IEEE Trans. Ind. Electron.*, vol. 70, no. 6, pp. 5797–5807, Jun. 2023.
[6] A. Prakash, P. Singh, K. Kumar, and S. Parida, "Design of a reduced-order WADC for wind turbine system-integrated power system," *IEEE Trans. Ind. Appl.*, vol. 58, no. 3, pp. 3250–3260, May/Jun. 2022.
[7] T. L. Van, T. H. Nguyen, and D.-C. Lee, "Advanced pitch angle control based on fuzzy logic for variable-speed wind turbine systems," *IEEE Trans. Energy Convers.*, vol. 30, no. 2, pp. 578–587, Jun. 2015.
[8] S. Hwang, J. B. Park, and Y. H. Joo, "Disturbance observer-based integral fuzzy sliding-mode control and its application to wind turbine system," *IET Control Theory Appl.*, vol. 13, pp. 1891–1900, Jul. 2019.

Authorized licensed use limited to: Institut Teknologi Sepuluh Nopember. Downloaded on March 29,2026 at 17:29:57 UTC from IEEE Xplore. Restrictions apply.

[9] A. Dida, F. Merahi, and S. Mekhilef, "New grid synchronization and power control scheme of doubly-fed induction generator based wind turbine system using fuzzy logic control," *Comput. Elect. Eng.*, vol. 84, Jun. 2020, Art. no. 106647.

[10] L. Shanmugam and Y. H. Joo, "Stabilization of permanent magnet synchronous generator-based wind turbine system via fuzzy-based sampled-data control approach," *Inf. Sci.*, vol. 559, pp. 270–285, Jun. 2021.

[11] P. Cheng, H. Wang, V. Stojanovic, F. Liu, S. He, and K. Shi, "Dissipativity-based finite-time asynchronous output feedback control for wind turbine system via a hidden Markov model," *Int. J. Syst. Sci.*, vol. 53, pp. 3177–3189, May 2022.

[12] L. Qi, H. Cao, Y. Zhang, M. Zhang, K. Shi, and J. Xu, "Data-driven control of the coupled loads for floating wind turbine systems based on deformable trailing edge flaps," *Ocean Eng.*, vol. 250, Apr. 2022, Art. no. 111083.

[13] P. Anbalagan and Y. H. Joo, "Dissipative-based sampled-data control for T-S fuzzy wind turbine system via fragmented-delayed state looped functional approach," *Nonlinear Dyn.*, vol. 111, pp. 2463–2486, Dec. 2022.

[14] X.-M. Zhang, Q.-L. Han, and X. Yu, "Survey on recent advances in networked control systems," *IEEE Trans. Ind. Inform.*, vol. 12, no. 5, pp. 1740–1752, Oct. 2016.

[15] M. N. Kurt, Y. Yilmaz, and X. Wang, "Real-time detection of hybrid and stealthy cyber-attacks in smart grid," *IEEE Trans. Inf. Forensics Secur.*, vol. 14, no. 2, pp. 498–513, Feb. 2019.

[16] Z. Cheng, S. Hu, D. Yue, C. Dou, and S. Shen, "Resilient distributed coordination control of multiarea power systems under hybrid attacks," *IEEE Trans. Syst., Man, Cybern. Syst.*, vol. 52, no. 1, pp. 7–18, Jan. 2022.

[17] Z. Lian, P. Shi, C.-C. Lim, and X. Yuan, "Fuzzy-model-based lateral control for networked autonomous vehicle systems under hybrid cyber-attacks," *IEEE Trans. Cybern.*, vol. 53, no. 4, pp. 2600–2609, Apr. 2023.

[18] S. Dong and M. Liu, "Adaptive fuzzy asynchronous control for nonhomogeneous Markov jump power systems under hybrid attacks," *IEEE Trans. Fuzzy Syst.*, vol. 31, no. 3, pp. 1009–1019, Mar. 2023.

[19] P. Tabuada, "Event-triggered real-time scheduling of stabilizing control tasks," *IEEE Trans. Autom. Control*, vol. 52, no. 9, pp. 1680–1685, Sep. 2007.

[20] D. Yue, E. Tian, and Q.-L. Han, "A delay system method for designing event-triggered controllers of networked control systems," *IEEE Trans. Autom. Control*, vol. 58, no. 2, pp. 475–481, Feb. 2013.

[21] Y. Fan, G. Feng, Y. Wang, and C. Song, "Distributed event-triggered control of multi-agent systems with combinational measurements," *Automatica*, vol. 49, pp. 671–675, Feb. 2013.

[22] A. Girard, "Dynamic triggering mechanisms for event-triggered control," *IEEE Trans. Autom. Control*, vol. 60, no. 7, pp. 1992–1997, Jul. 2015.

[23] R. Yang, H. Zhang, G. Feng, H. Yan, and Z. Wang, "Robust cooperative output regulation of multi-agent systems via adaptive event-triggered control," *Automatica*, vol. 102, pp. 129–136, Apr. 2019.

[24] E. Tian and C. Peng, "Memory-based event-triggering H_∞ load frequency control for power systems under deception attacks," *IEEE Trans. Cybern.*, vol. 50, no. 11, pp. 4610–4618, Nov. 2020.

[25] P. Mani and Y. H. Joo, "Fuzzy event-triggered control for back-to-back converter involved PMSG-based wind turbine systems," *IEEE Trans. Fuzzy Syst.*, vol. 30, no. 5, pp. 1409–1420, May 2022.

[26] S. Yan, Z. Gu, J. H. Park, and X. Xie, "Adaptive memory-event-triggered static output control of T–S fuzzy wind turbine systems," *IEEE Trans. Fuzzy Syst.*, vol. 30, no. 9, pp. 3894–3904, Sep. 2022.

[27] V. Gandhi and Y. H. Joo, "T–S fuzzy sampled-data control for nonlinear systems with actuator faults and its application to wind energy system," *IEEE Trans. Fuzzy Syst.*, vol. 30, no. 2, pp. 462–474, Feb. 2020.

[28] Y. Liu, Z. Fang, J. H. Park, and F. Fang, "Quantized event-triggered synchronization of discrete-time chaotic neural networks with stochastic deception attack," *IEEE Trans. Syst., Man, Cybern. Syst.*, vol. 53, no. 7, pp. 4511–4521, Jul. 2023.

[29] J. Zhou, D. Xu, W. Tai, and C. K. Ahn, "Switched event-triggered H_∞ security control for networked systems vulnerable to aperiodic DoS attacks," *IEEE Trans. Netw. Sci. Eng.*, vol. 10, no. 4, pp. 2109–2123, Jul./Aug. 2023.

[30] P. Chen, D. Zhang, L. Yu, and H. Yan, "Dynamic event-triggered output feedback control for load frequency control in power systems with multiple cyber attacks," *IEEE Trans. Syst., Man, Cybern. Syst.*, vol. 52, no. 10, pp. 6246–6258, Oct. 2022.

[31] A. Wang, M. Fei, Y. Song, C. Peng, D. Du, and Q. Sun, "Secure adaptive event-triggered control for cyber–physical power systems under denial-of-service attacks," *IEEE Trans. Cybern.*, vol. 54, no. 3, pp. 1722–1733, Mar. 2024.

[32] Z. Hu, S. Liu, W. Luo, and L. Wu, "Resilient distributed fuzzy load frequency regulation for power systems under cross-layer random denial-of-service attacks," *IEEE Trans. Cybern.*, vol. 52, no. 4, pp. 2396–2406, Apr. 2022.

[33] W. Yu, R. Wang, X. Bu, Z. Hou, and Z. Wu, "Resilient model-free adaptive iterative learning control for nonlinear systems under periodic DoS attacks via a fading channel," *IEEE Trans. Syst., Man, Cybern. Syst.*, vol. 52, no. 7, pp. 4117–4128, Jul. 2022.

[34] M. Rathee, S. Kumar, A. H. Gandomi, K. Dilip, B. Balusamy, and R. Patan, "Ant colony optimization based quality of service aware energy balancing secure routing algorithm for wireless sensor networks," *IEEE Trans. Eng. Manage.*, vol. 68, no. 1, pp. 170–182, Feb. 2021.

[35] Y. Pan, Y. Wu, and H.-K. Lam, "Security-based fuzzy control for nonlinear networked control systems with DoS attacks via a resilient event-triggered scheme," *IEEE Trans. Fuzzy Syst.*, vol. 30, no. 10, pp. 4359–4368, Oct. 2022.

[36] D. Yue, E. Tian, Y. Zhang, and C. Peng, "Delay-distribution-dependent robust stability of uncertain systems with time-varying delay," *Int. J. Robust Nonlinear Control: IFAC-Affiliated J.*, vol. 19, pp. 377–393, Apr. 2009.

[37] H. Khalil, *Nonlinear Systems*, 3rd ed., Hoboken, NJ, USA: Prentice-Hall, 2002.

[38] P. Park, J. W. Ko, and C. Jeong, "Reciprocally convex approach to stability of systems with time-varying delays," *Automatica*, vol. 47, pp. 235–238, Jan. 2011.

Authorized licensed use limited to: Institut Teknologi Sepuluh Nopember. Downloaded on March 29,2026 at 17:29:57 UTC from IEEE Xplore. Restrictions apply.